

Modeling Linear and Integer Programs

Part 2b: Integer Programs

IE 311 – Operations Research I

Recall: Types of Linear Optimization Models

A **Linear Program** (continuous variables):

$$\max; 2x_1 + 5x_2 \quad \text{s.t.} \quad x_1 + 2x_2 \leq 5, ; ; x_1, x_2 \geq 0$$

A **(Pure) Integer (Linear) Program** (integer variables):

$$\max; 2x_1 + 5x_2 \quad \text{s.t.} \quad x_1 + 2x_2 \leq 5, ; ; x_1, x_2 \geq 0 \text{ and } x_1, x_2 \text{ integer}$$

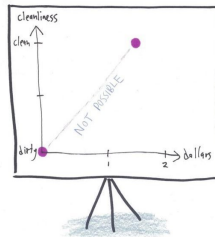
A **Mixed-Integer Linear Program** (continuous and integer variables):

$$\max; 2x_1 + 5x_2 \quad \text{s.t.} \quad x_1 + 2x_2 \leq 5, ; ; x_1, x_2 \geq 0 \text{ and } x_1 \text{ integer}$$

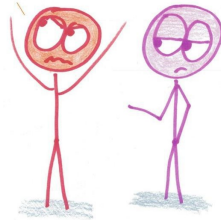
Is Integer Programming Really Needed?



Um, laundry machines only perform integer numbers of cycles.



I HATE
DISCRETE MATH



Integer Programming is needed when

- the continuity assumption on the decision variables is not valid, or
- the following types of decisions are to be modeled:
- "yes/no" decisions,
- logical statements such as "either-or", "if-then", "and", "exclusive-or".

We will see examples of IP formulations and how to implement and solve them in Python–Gurobi. The solution methods for IP models are covered in IE 312.

A First Example: Knapsack IP

- Suppose there is a knapsack of capacity 14 units.
- There are four items, each with a size and a value:

	Item 1	Item 2	Item 3	Item 4
Value	5	7	4	3
Size	8	11	6	4

- Determine which items to pick to maximize the value while respecting the capacity.

Decision variables: $x_i = 1$ if item i is picked and 0 otherwise.

Objective – maximize the total value; **constraints** – knapsack capacity and binary requirements:

$$\begin{array}{ll}\max & 5x_1 + 7x_2 + 4x_3 + 3x_4 \quad (\text{value}) \\ \text{s.t.} & 8x_1 + 11x_2 + 6x_3 + 4x_4 \leq 14 \quad (\text{capacity}) \\ & x_1, x_2, x_3, x_4 \in 0, 1 \quad (\text{binary})\end{array}$$

Knapsack IP: Simplest (but Dirty) Python–Gurobi Implementation

```
from gurobipy import GRB, Model
model = Model('Knapsack')
x1 = model.addVar(vtype=GRB.BINARY)    # binary variable x1
x2 = model.addVar(vtype=GRB.BINARY)    # binary variable x2
x3 = model.addVar(vtype=GRB.BINARY)    # binary variable x3
x4 = model.addVar(vtype=GRB.BINARY)    # binary variable x4
model.setObjective(5*x1+7*x2+4*x3+3*x4, GRB.MAXIMIZE) # maximize value
model.addConstr(8*x1+11*x2+6*x3+4*x4 <= 14)          # capacity
model.optimize()                                     # optimize
```

- Stockco is considering four investments with net present values (NPV) \$16,000, \$22,000, \$12,000, \$8,000.
- They require present cash outflows of \$5,000, \$7,000, \$4,000, \$3,000.
- Currently \$14,000 is available for investment.
- Formulate an IP to maximize the NPV obtained from investments 1–4, then add:
- Stockco can invest in at most two investments.
- If it invests in investment 2, it must also invest in investment 1.
- If it invests in investment 2, it cannot invest in investment 4.

$x_j = 1$ if investment j is selected and 0 otherwise.

$$\begin{array}{ll}\max & 16x_1 + 22x_2 + 12x_3 + 8x_4 \quad (\text{maximize NPV}) \\ \text{s.t.} & 5x_1 + 7x_2 + 4x_3 + 3x_4 \leq 14 \quad (\text{available budget}) \\ & x_1, x_2, x_3, x_4 \in 0, 1 \quad (\text{binary})\end{array}$$

(Budget and NPVs in thousands of dollars.)

- Stockco can invest in at most two investments:

$$x_1 + x_2 + x_3 + x_4 \leq 2$$

- If it invests in investment 2, it must also invest in investment 1:

$$(x_2 = 1 \Rightarrow x_1 = 1) \iff x_2 \leq x_1$$

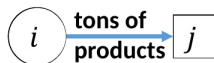
- If it invests in investment 2, it cannot invest in investment 4:

$$(x_2 = 1 \Rightarrow x_4 = 0) \iff x_2 \leq 1 - x_4$$

- A company is operating m supply stations to satisfy the demand of n demand centers.
- Each supply station has a capacity of at most s_i tons of products and each demand center requires at least d_j tons of products (you can model the amount of product as a continuous variable).
- The products are shipped by using trucks. Each truck can carry at most b tons of products (you need to model the number of trucks as an integer variable).
- The cost of sending a truck from supply station i to demand center j is c_{ij} , regardless of the amount of product carried by the truck.
- Write a mixed integer linear program to minimize the total cost.

- **Decision variables** (what can be decided/changed?)

- We need to know **how many tons of products** are sent from each supply station i to each demand center j .



- We also need to know **how many trucks** are sent from each supply station i to each demand center j .



- Note the following relationship:

$$\text{\#trucks} = \left\lceil \frac{\text{tons of products}}{\text{truck capacity}} \right\rceil$$

Distribution Problem: Model

x_{ij} = number of trucks sent from supply station i to demand center j ; y_{ij} = tons of product sent from i to j . Each truck has capacity b , so $x_{ij} \geq y_{ij}/b$.

$$\begin{aligned} \min \quad & \sum_{i=1}^m \sum_{j=1}^n c_{ij} x_{ij} && \text{(total cost)} \\ \text{s.t.} \quad & \sum_{j=1}^n y_{ij} \leq s_i, \quad i = 1, \dots, m && \text{(capacity } i) \\ & \sum_{i=1}^m y_{ij} \geq d_j, \quad j = 1, \dots, n && \text{(demand } j) \\ & x_{ij} \geq \frac{y_{ij}}{b}, \quad i, j && \text{(truck count)} \\ & x_{ij} \geq 0 \text{ and integer, } y_{ij} \geq 0 \end{aligned}$$

- A machine tool plant owns m different machines on which it can process n types of jobs.
- A job j takes t_{ij} time units if processed on machine i .
- A job cannot be divided between machines, that is, each job must be processed by exactly one machine.
- Each machine can process more than one job.
- If a machine is used at all, then a setup time of s_i units is needed.
- Formulate an IP to minimize the sum of total processing and setup times.

- **Decision variables** (what can be decided/changed?)
 - We need to know **job j is processed on which machine i .**
 - We also need to know **which machine i is used at all.**
 - Note the following relationship:
 - **If a machine i is NOT used at all, then machine i CANNOT process job j .**

Machine Allocation Problem: Model

$x_{ij} = 1$ if machine i processes job j (else 0); $y_i = 1$ if machine i is assigned at least one job (else 0). Relationship: $y_i = 0 \Rightarrow x_{ij} = 0$.

$$\begin{aligned} \min \quad & \sum_{i=1}^m \sum_{j=1}^n t_{ij} x_{ij} + \sum_{i=1}^m s_i y_i && \text{(total time)} \\ \text{s.t.} \quad & \sum_{i=1}^m x_{ij} = 1, \quad j = 1, \dots, n && \text{(each job assigned)} \\ & x_{ij} \leq y_i, \quad i = 1, \dots, m, \quad j = 1, \dots, n && \text{(machine setup if used)} \\ & x_{ij}, y_i \in 0, 1 \end{aligned}$$

There are six cities in Kilroy County. Determine the minimum number of fire stations so that at least one station is within 15 minutes' driving time of each city. Driving times (minutes) between cities:

	City 1	City 2	City 3	City 4	City 5	City 6
City 1	0	10	20	30	30	20
City 2	10	0	25	35	20	10
City 3	20	25	0	15	30	20
City 4	30	35	15	0	15	25
City 5	30	20	30	15	0	14
City 6	20	10	20	25	14	0

Set Covering: Model

$x_j = 1$ if a fire station is built in city j (else 0); d_{ij} = distance between cities i and j ; T = distance threshold (15 min).

$$\begin{aligned} \min \quad & \sum_{j=1}^6 x_j && \text{(minimize the number of stations)} \\ \text{s.t.} \quad & \sum_{j: d_{ij} \leq T} x_j \geq 1, \quad i = 1, \dots, 6 && \text{(coverage)} \\ & x_j \in 0, 1, \quad j = 1, \dots, 6 && \text{(binary)} \end{aligned}$$

By building a station in City 1, for example, we cover cities 1 and 2.

Gandhi Cloth Co. can make shirts, shorts, and pants. Making each requires renting the appropriate machinery (\$200, \$150, \$100 per week). Each week 150 labor hours and 160 sq yd of cloth are available.

Clothing	Labor (hrs)	Cloth (yd)	Price (\$)	Cost (\$)
Shirt	3	4	12	6
Shorts	2	3	8	4
Pants	6	4	15	8

Formulate an IP to maximize profit (production amounts are integer).

x_j = amount of cloth type j produced; $y_j = 1$ if type j is produced (else 0), $j = 1, 2, 3$.

$$\begin{aligned} \max \quad & (12 - 6)x_1 + (8 - 4)x_2 + (15 - 8)x_3 - 200y_1 - 150y_2 - 100y_3 \\ \text{s.t.} \quad & 3x_1 + 2x_2 + 6x_3 \leq 150 && \text{(labor)} \\ & 4x_1 + 3x_2 + 4x_3 \leq 160 && \text{(cloth)} \\ & (y_j = 0 \Rightarrow x_j = 0) \iff x_j \leq M_j y_j, \quad j = 1, 2, 3 \\ & x_1, x_2, x_3 \geq 0 \text{ and integer, } y_1, y_2, y_3 \in 0, 1 \end{aligned}$$

Fixed Charge: General Model

Parameters: c_j, p_j = unit cost and selling price of type j ; r_j = machine rental cost for type j ;
 l_j, a_j = unit labor and cloth requirements; L, A = total labor and cloth availability;
 $M_j = \min \lfloor L/l_j \rfloor, \lfloor A/a_j \rfloor, j = 1, \dots, n$.

$$\max \sum_{j=1}^n [(p_j - c_j)x_j - r_j y_j] \quad (\text{maximize profit})$$

$$\text{s.t.} \quad \sum_{j=1}^n l_j x_j \leq L \quad (\text{available labor})$$

$$\sum_{j=1}^n a_j x_j \leq A \quad (\text{available cloth})$$

$$x_j \leq M_j y_j \quad (\text{big-M constraint})$$

$$x_j \geq 0 \text{ and integer, } y_j \in \{0, 1\}, j = 1, \dots, n$$

A central use of binary variables is to encode logical conditions – "either-or", "if-then", "and", and "exclusive-or" – as linear constraints. The tables that follow give the standard linearizations.

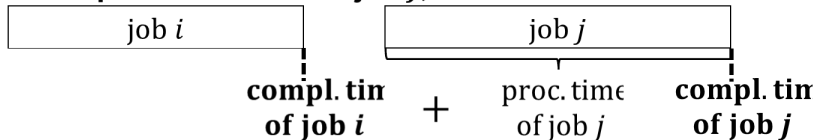
- Let x, y, z be **binary** variables. Let us see how to model some logical relationships:
- $z = x \text{ and } y$ can be modelled as $x + y - 1 \leq z, z \leq x, z \leq y$.
- $z = x \text{ or } y$ can be modelled as $x + y \geq z, z \geq x, z \geq y$.
- $z = x \text{ xor } y$ can be modelled as $x + y = z + 2w, w \in \{0,1\}$.
- $y = \text{not } x$ can be modelled as $y = 1 - x$.
- $x \rightarrow y$ can be modelled as $x \leq y$.
- $x \rightarrow \text{not } y$ can be modelled as $x \leq 1 - y$.

- Let $f_1(x_1, \dots, x_n), f_2(x_1, \dots, x_n)$ be **functions** of decision variables $x_1, \dots, x_n$.
- We will assume that there exist two constants M_1, M_2 which upper bound f_1, f_2 for every value of $x_1, \dots, x_n$.
- Let y be a **binary** variable. Let us see how to model some logical relationships:
- $f_1(x_1, \dots, x_n) \leq 0$ or $f_2(x_1, \dots, x_n) \leq 0$ can be modelled as $f_1(x_1, \dots, x_n) \leq M_1 y, f_2(x_1, \dots, x_n) \leq M_2(1 - y)$.
- $f_1(x_1, \dots, x_n) > 0 \rightarrow f_2(x_1, \dots, x_n) \leq 0$ can be modelled as $f_1(x_1, \dots, x_n) \leq M_1 y, f_2(x_1, \dots, x_n) \leq M_2(1 - y)$.

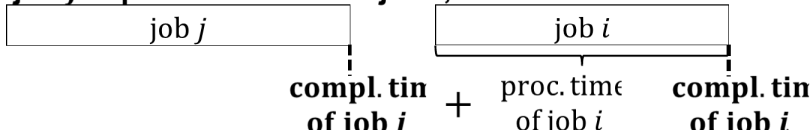
- Jobs $1, \dots, n$ must be processed on a single machine.
- Each job is available for processing after a certain time, called release time.
- For each job i , we are given
 - its release time r_i ,
 - its processing time p_i and
 - its weight w_i .
- Formulate as a mixed-integer linear program the problem of sequencing the jobs without overlap or interruption so that the sum of the weighted completion times is minimized.

Decision variables (what can be decided/changed?)

- We need to know **the completion time of each job i** .
- We also need to know **whether job i is processed before job j or not**.
- If job i is processed before job j , then we must have



- If job j is processed before job i , then we must have



C_i = completion time of job i ($i = 1, \dots, n$); $x_{ij} = 1$ if job i is processed before job j (else 0), $1 \leq i < j \leq n$.

$$\min \sum_{i=1}^n w_i C_i \quad (\text{weighted completion time})$$

$$\text{s.t. } C_i \geq r_i + p_i, \quad i = 1, \dots, n \quad (\text{lower bound})$$

$$C_j \geq C_i + p_j - M(1 - x_{ij}), \quad 1 \leq i < j \leq n \quad (i \text{ before } j)$$

$$C_i \geq C_j + p_i - Mx_{ij}, \quad 1 \leq i < j \leq n \quad (j \text{ before } i)$$

$$C_i \geq 0, \quad x_{ij} \in 0, 1$$

A valid big- M is $M = \sum_{i=1}^n p_i + \max_i r_i$.

Euing Gas produces gas 1 and gas 2 from oil 1 and oil 2. Each gallon of gas 1 must contain $\geq 50\%$ oil 1; each gallon of gas 2, $\geq 60\%$ oil 1. Gas 1 sells for 12¢/gal, gas 2 for 14¢/gal. Currently 500 gal of oil 1 and 1,000 gal of oil 2 are available. Up to 1,500 more gal of oil 1 can be bought: first 500 at 25¢, next 500 at 20¢, next 500 at 15¢ per gallon. Formulate an IP to maximize Euing's profit.

x_{ij} = gallons of oil i used in gas j ($i, j = 1, 2$); z = gallons of oil 1 purchased; c = total cost of purchasing oil 1.

$$\begin{array}{ll}\max & 12(x_{11} + x_{21}) + 14(x_{12} + x_{22}) - c \\ \text{s.t.} & x_{11} \geq 0.5(x_{11} + x_{21}) \quad (\text{oil 1 in gas 1}) \\ & x_{12} \geq 0.6(x_{12} + x_{22}) \quad (\text{oil 1 in gas 2}) \\ & x_{11} + x_{12} \leq 500 + z \quad (\text{oil 1 availability}) \\ & x_{21} + x_{22} \leq 1000 \quad (\text{oil 2 availability}) \\ & x_{11}, x_{12}, x_{21}, x_{22}, z \geq 0\end{array}$$

The purchasing cost is the (nonconvex) piecewise-linear function

$$c = \begin{cases} 25z & 0 \leq z \leq 500 \\ 12500 + 20(z - 500) & 500 \leq z \leq 1000 \\ 22500 + 15(z - 1000) & 1000 \leq z \leq 1500 \end{cases}$$

Minimizing a nonconvex piecewise-linear function is "difficult" – so we linearize it using binary variables.

$$\begin{aligned} \max \quad & 12(x_{11} + x_{21}) + 14(x_{12} + x_{22}) - c \\ c = & (25z_1) + (20z_2 + 2500w_2) + (15z_3 + 7500w_3) \\ z = & z_1 + z_2 + z_3, \quad w_1 + w_2 + w_3 = 1 \\ 0 \leq & z_1 \leq 500w_1, \quad 500w_2 \leq z_2 \leq 1000w_2, \quad 1000w_3 \leq z_3 \leq 1500w_3 \\ x_{11} \geq & 0.5(x_{11} + x_{21}), \quad x_{12} \geq 0.6(x_{12} + x_{22}) \\ x_{11} + & x_{12} \leq 500 + (z_1 + z_2 + z_3), \quad x_{21} + x_{22} \leq 1000 \\ x_{ij} \geq & 0, \quad w_1, w_2, w_3 \in 0, 1 \end{aligned}$$

Dorian Auto may make compact, midsize, and large cars. Currently 6,000 tons of steel and 60,000 labor hours are available. For a type to be economical, at least 1,000 of that type must be produced.

Resource	Compact	Midsize	Large
Steel required	1.5 tons	3 tons	5 tons
Labor required	30 hrs	25 hrs	40 hrs
Profit yielded	\$2,000	\$3,000	\$4,000

x_j = number of type j cars produced; $y_j = 1$ if type j is produced (else 0), $j = 1, 2, 3$.

$$\begin{array}{ll}\max & 2000x_1 + 3000x_2 + 4000x_3 \\ \text{s.t.} & 30x_1 + 25x_2 + 40x_3 \leq 60000 \quad (\text{labor}) \\ & 1.5x_1 + 3x_2 + 5x_3 \leq 6000 \quad (\text{steel}) \\ & 1000y_j \leq x_j \leq M_j y_j, \quad j = 1, 2, 3 \quad (\text{either–or}) \\ & x_1, x_2, x_3 \geq 0, \quad y_1, y_2, y_3 \in 0, 1\end{array}$$

Box Problem: "And"

A company sells seven box types (volumes 17–33 cubic feet). Producing a box costs its volume; a fixed cost of \$1000 is incurred to produce any of a type. All demand must be met; a demand may be met by a box of larger size. If both box 5 and box 7 are made, the combined fixed cost is \$1500 (not \$2000). At least 5 types must be made. Formulate a MILP to minimize total cost.

	Box 1	Box 2	Box 3	Box 4	Box 5	Box 6	Box 7
Size	33	30	26	24	19	18	17
Demand	400	300	500	700	200	400	200

Box Problem: Variables and Objective

s_i = size of box i ; d_i = demand of box i ($i = 1, \dots, 7$); x_i = amount of box i produced; $y_i = 1$ if box i is produced (else 0); z_{ij} = amount of box i used to satisfy demand of box j ; $w = 1$ if both box 5 and box 7 are produced (else 0).

$$\min; \sum_{i=1}^7 s_i x_i + 1000 \sum_{i=1}^7 y_i - 500w$$

$$\sum_{j=i}^7 z_{ij} = x_i, \quad i = 1, \dots, 7 \quad (\text{box } i \text{ produced})$$

$$\sum_{i=1}^j z_{ij} = d_j, \quad j = 1, \dots, 7 \quad (\text{demand } j \text{ met})$$

$$x_i \leq M y_i, \quad i = 1, \dots, 7 \quad (\text{fixed charge})$$

$$y_5 + y_7 - 1 \leq w, \quad w \leq y_5, \quad w \leq y_7 \quad (\text{"and"})$$

$$\sum_{i=1}^7 y_i \geq 5 \quad (\text{at least 5 types})$$

$$x_i, z_{ij} \geq 0, \quad y_i, w \in \{0, 1\}; \quad z_{ij} = 0 \text{ for } 1 \leq j < i \leq 7$$

Determine a seating arrangement of 8 diners – 4 girls (A, B, C, D) and 4 boys (E, F, G, H) – around a circular table such that:

- no boy is seated next to a boy;
- A and E are not seated next to each other;
- C and H are seated next to each other;
- if A and F are seated next to each other, then D and F must be next to each other.

Formulate an integer linear program to determine whether a feasible arrangement exists.

$x_{ij} = 1$ if child i sits in seat j (else 0), $i = A, \dots, H$, $j = 1, \dots, 8$ (seats arranged circularly).

$$\sum_{i=A}^H x_{ij} = 1, \quad j = 1, \dots, 8 \quad (\text{one child in seat } j)$$

$$\sum_{j=1}^8 x_{ij} = 1, \quad i = A, \dots, H \quad (\text{child } i \text{ in one seat})$$

No boy next to a boy ($i, k = E, \dots, H$):

$$x_{i,j} + x_{k,j+1} \leq 1, \quad j = 1, \dots, 7; \quad x_{i,8} + x_{k,1} \leq 1.$$

A and E not adjacent ($j = 2, \dots, 7$):

$$x_{A,j} + x_{E,j+1} \leq 1, \quad x_{A,j} + x_{E,j-1} \leq 1; \quad x_{A,8} + x_{E,1} \leq 1, \quad x_{A,1} + x_{E,8} \leq 1.$$

C and H adjacent ($j = 2, \dots, 7$):

$$x_{H,j-1} + x_{H,j+1} \geq x_{C,j}; \quad x_{H,8} + x_{H,2} \geq x_{C,1}, \quad x_{H,7} + x_{H,1} \geq x_{C,8}.$$

If A and F are adjacent, then D and F must be adjacent:

$$x_{D,j+2} \geq x_{A,j} + x_{F,j+1} - 1, \quad j = 1, \dots, 6$$

$$x_{D,2} \geq x_{A,8} + x_{F,1} - 1, \quad x_{D,1} \geq x_{A,7} + x_{F,8} - 1$$

$$x_{D,j-2} \geq x_{A,j} + x_{F,j-1} - 1, \quad j = 3, \dots, 8$$

$$x_{D,7} \geq x_{A,1} + x_{F,8} - 1, \quad x_{D,8} \geq x_{A,2} + x_{F,1} - 1$$